



PREMIUM

Full-Length Mock Test

Mock 01

Chemical Engineering

GATE CH

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Process Calculations — Stoichiometry]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.2 [1 Mark] [EASY] [Mass Transfer — Humidification]

For CSTR, conversion X vs space time τ :

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.3 [1 Mark] [EASY] [Heat Transfer — Conduction]

Prandtl number Pr =:

- A) $\frac{\rho C_p k}{\mu}$
- B) $\frac{\mu}{\rho C_p k}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{1}{\rho C_p k}$

Q.4 [1 Mark] [EASY] [CRE — CSTR/PFR]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.5 [1 Mark] [EASY] [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ :

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.6 [1 Mark] [EASY] [Process Calculations — Energy Balance]

Prandtl number Pr =:

- A) $\frac{\rho C_p k}{\mu}$
- B) $\frac{\mu}{\rho C_p k}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{1}{\rho C_p k}$

Q.7 [1 Mark] [EASY] [Mass Transfer — Drying]

Bubble cap tray is:

A) Tray type

B) Packing

C) Valve

D) Sieve

Q.8 [1 Mark] [EASY] [Heat Transfer — Heat Exchangers]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.9 [1 Mark] [EASY] [CRE — CSTR/PFR]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho c_p k}{\mu}$
- C) $\frac{\mu}{\rho c_p k}$
- D) $\frac{\rho c_p \mu}{k}$

Q.10 [1 Mark] [EASY] [Fluid Mechanics — Non-Newtonian]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.11 [1 Mark] [EASY] [Process Calculations — Bypass]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.12 [1 Mark] [EASY] [Mass Transfer — Extraction]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho c_p k}{\mu}$
- C) $\frac{\mu}{\rho c_p k}$
- D) $\frac{\rho c_p \mu}{k}$

Q.13 [2 Marks] [MODERATE] [Heat Transfer — Boiling]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.14 [2 Marks] [MODERATE] [CRE — Rate Laws]

For CSTR, conversion X vs space time τ

- A) Linear

B) Exponential

C) Hyperbolic

D) Parabolic

Q.15 [2 Marks] [MODERATE] [Fluid Mechanics — Laminar/Turbulent]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho k}{\mu}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{1}{Pr}$

Q.16 [2 Marks] [MODERATE] [Process Calculations — Bypass]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.17 [2 Marks] [MODERATE] [Mass Transfer — Distillation]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.18 [2 Marks] [MODERATE] [Heat Transfer — Convection]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho k}{\mu}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{1}{Pr}$

Q.19 [2 Marks] [MODERATE] [CRE — Rate Laws]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.20 [2 Marks] [MODERATE] [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.21 [2 Marks] [MODERATE] [Process Calculations — Material Balance]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$

B) ; $6\frac{1}{2}$

C) ; $\dot{Y}_S \pm$

D) $1/(\dot{\phi} \pm)$

Q.22 [2 Marks] [MODERATE] [Mass Transfer — Drying]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.23 [2 Marks] [MODERATE] [Heat Transfer — Radiation]

For CSTR, conversion X vs space time τ :

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.24 [2 Marks] [MODERATE] [CRE — Rate Laws]

Prandtl number Pr =:

- A) $\frac{\mu}{\rho k_f}$
- B) $\frac{\mu}{\rho C_p}$
- C) $\frac{\mu}{\rho k_f}$
- D) $\frac{1}{(\frac{\mu}{\rho k_f})}$

Q.25 [2 Marks] [MODERATE] [Fluid Mechanics — Compressible Flow]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.26 [2 Marks] [MODERATE] [Process Calculations — Energy Balance]

For CSTR, conversion X vs space time τ :

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.27 [2 Marks] [MODERATE] [Mass Transfer — Absorption]

Prandtl number Pr =:

- A) $\frac{\mu}{\rho k_f}$
- B) $\frac{\mu}{\rho C_p}$
- C) $\frac{\mu}{\rho k_f}$
- D) $\frac{1}{(\frac{\mu}{\rho k_f})}$

Q.28 [2 Marks] [MODERATE] [Heat Transfer — Heat Exchangers]

Bubble cap tray is:

- A) Tray type

B) Packing

C) Valve

D) Sieve

Q.29 [2 Marks] [MODERATE] [CRE — CSTR/PFR]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.30 [2 Marks] [MODERATE] [Fluid Mechanics — Laminar/Turbulent]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho k}{\mu}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{\rho C_p}{\mu}$

Q.31 [2 Marks] [MODERATE] [Process Calculations — Energy Balance]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.32 [2 Marks] [MODERATE] [Mass Transfer — Drying]

For CSTR, conversion X vs space time τ

- A) Linear
- B) Exponential
- C) Hyperbolic
- D) Parabolic

Q.33 [2 Marks] [HARD] [Heat Transfer — Boiling]

Prandtl number $Pr =$:

- A) $\frac{\mu}{\rho k}$
- B) $\frac{\rho k}{\mu}$
- C) $\frac{\mu}{\rho C_p}$
- D) $\frac{\rho C_p}{\mu}$

Q.34 [2 Marks] [HARD] [CRE — Reactor Design]

Bubble cap tray is:

- A) Tray type
- B) Packing
- C) Valve
- D) Sieve

Q.35 [2 Marks] [HARD] [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ

- A) Linear

B) Exponential

C) Hyperbolic

D) Parabolic

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Process Calculations — Recycle]

Which are non-Newtonian fluids?

- A) Water
 - B) Blood
 - C) Ketchup
 - D) Honey
 - E) Toothpaste
-

Q.37 [2 Marks] [HARD] [Mass Transfer — Absorption]

Which are non-Newtonian fluids?

- A) Water
 - B) Blood
 - C) Ketchup
 - D) Honey
 - E) Toothpaste
-

Q.38 [2 Marks] [HARD] [Heat Transfer — Heat Exchangers]

Which are non-Newtonian fluids?

- A) Water
 - B) Blood
 - C) Ketchup
 - D) Honey
 - E) Toothpaste
-

Q.39 [2 Marks] [HARD] [CRE — Reactor Design]

Which are non-Newtonian fluids?

- A) Water
 - B) Blood
 - C) Ketchup
 - D) Honey
 - E) Toothpaste
-

Q.40 [2 Marks] [HARD] [Fluid Mechanics — Non-Newtonian]

Which are non-Newtonian fluids?

- A) Water
 - B) Blood
 - C) Ketchup
 - D) Honey
 - E) Toothpaste
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Process Calculations — Material Balance]**

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time τ

Answer: _____

Q.52 [2 Marks] [HARD] [Mass Transfer — Extraction]

Distillation: relative volatility α ; Minimum reflux ratio $R_{\min}$ at $\alpha = 2$ and $x_D = 0.9$

Answer: _____

Q.53 [2 Marks] [HARD] [Heat Transfer — Heat Exchangers]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time τ

Answer: _____

Q.54 [2 Marks] [HARD] [CRE — CSTR/PER]

Distillation: relative volatility α ; Minimum reflux ratio $R_{\min}$ at $\alpha = 2$ and $x_D = 0.9$

Answer: _____

Q.55 [2 Marks] [HARD] [Fluid Mechanics — Non-Newtonian]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time τ

Answer: _____

Q.56 [2 Marks] [HARD] [Process Calculations — Stoichiometry]

Distillation: relative volatility α ; Minimum reflux ratio $R_{\min}$ at $\alpha = 2$ and $x_D = 0.9$

Answer: _____

Q.57 [2 Marks] [HARD] [Mass Transfer — Distillation]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time τ

Answer: _____

Q.58 [2 Marks] [HARD] [Heat Transfer — Heat Exchangers]

Distillation: relative volatility α ; Minimum reflux ratio $R_{\min}$ at $\alpha = 2$ and $x_D = 0.9$

Answer: _____

Answer: _____

Q.59 [2 Marks] [HARD] [CRE — Homogeneous/Heterogeneous]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time τ

Answer: _____

Q.60 [2 Marks] [HARD] [Fluid Mechanics — Non-Newtonian]

Distillation: relative volatility α ; Minimum reflux ratio at $\alpha = 1$

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	A	Q.2:	A	Q.3:	A
Q.4:	A	Q.5:	A	Q.6:	A
Q.7:	A	Q.8:	A	Q.9:	A
Q.10:	A	Q.11:	A	Q.12:	A
Q.13:	A	Q.14:	A	Q.15:	A
Q.16:	A	Q.17:	A	Q.18:	A
Q.19:	A	Q.20:	A	Q.21:	A
Q.22:	A	Q.23:	A	Q.24:	A
Q.25:	A	Q.26:	A	Q.27:	A
Q.28:	A	Q.29:	A	Q.30:	A
Q.31:	A	Q.32:	A	Q.33:	A
Q.34:	A	Q.35:	A	Q.36:	BCDE
Q.37:	BCDE	Q.38:	BCDE	Q.39:	BCDE
Q.40:	BCDE	Q.41:	BCDE	Q.42:	BCDE
Q.43:	BCDE	Q.44:	BCDE	Q.45:	BCDE
Q.46:	BCDE	Q.47:	BCDE	Q.48:	BCDE
Q.49:	BCDE	Q.50:	BCDE	Q.51:	10
Q.52:	2	Q.53:	10	Q.54:	2
Q.55:	10	Q.56:	2	Q.57:	10
Q.58:	2	Q.59:	10	Q.60:	2

Detailed Solutions

Question 1 [Process Calculations — Stoichiometry]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 2 [Mass Transfer — Humidification]

For CSTR, conversion X vs space time τ :

Answer: (A)

For first order in CSTR: $X = k\tau / (1 + k\tau)$, not linear. For zero order: linear.

Question 3 [Heat Transfer — Conduction]

Prandtl number Pr =:

Answer: (A)

$Pr = \mu C_p / k$ = Momentum diffusivity / thermal diffusivity.

Question 4 [CRE — CSTR/PFR]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 5 [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ :

Answer: (A)

For first order in CSTR: $X = k\tau / (1 + k\tau)$, not linear. For zero order: linear.

Question 6 [Process Calculations — Energy Balance]

Prandtl number Pr =:

Answer: (A)

$Pr = \mu C_p / k$ = Momentum diffusivity / thermal diffusivity.

Question 7 [Mass Transfer — Drying]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 8 [Heat Transfer — Heat Exchangers]

For CSTR, conversion X vs space time τ :

Answer: (A)

For first order in CSTR: $X = k\tau / (1 + k\tau)$, not linear. For zero order: linear.

Question 9 [CRE — CSTR/PFR]

Prandtl number Pr =:

Answer:

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 10 [Fluid Mechanics — Non-Newtonian]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 11 [Process Calculations — Bypass]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 12 [Mass Transfer — Extraction]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 13 [Heat Transfer — Boiling]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 14 [CRE — Rate Laws]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 15 [Fluid Mechanics — Laminar/Turbulent]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 16 [Process Calculations — Bypass]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 17 [Mass Transfer — Distillation]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 18 [Heat Transfer — Convection]

Prandtl number $Pr =$:

Answer:

$Pr = \frac{\mu}{\rho \alpha} = \frac{\mu}{\rho \frac{k}{\rho c_p}} = \frac{\mu c_p}{k}$ Momentum diffusivity / thermal diffusivity.

Question 19 [CRE — Rate Laws]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 20 [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 21 [Process Calculations — Material Balance]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{\rho \alpha} = \frac{\mu}{\rho \frac{k}{\rho c_p}} = \frac{\mu c_p}{k}$ Momentum diffusivity / thermal diffusivity.

Question 22 [Mass Transfer — Drying]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 23 [Heat Transfer — Radiation]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 24 [CRE — Rate Laws]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{\rho \alpha} = \frac{\mu}{\rho \frac{k}{\rho c_p}} = \frac{\mu c_p}{k}$ Momentum diffusivity / thermal diffusivity.

Question 25 [Fluid Mechanics — Compressible Flow]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 26 [Process Calculations — Energy Balance]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 27 [Mass Transfer — Absorption]

Prandtl number $Pr =$:

Answer:

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 28 [Heat Transfer — Heat Exchangers]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 29 [CRE — CSTR/PFR]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 30 [Fluid Mechanics — Laminar/Turbulent]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 31 [Process Calculations — Energy Balance]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 32 [Mass Transfer — Drying]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 33 [Heat Transfer — Boiling]

Prandtl number $Pr =$:

Answer: (A)

$Pr = \frac{\mu}{k} = \frac{\text{Momentum diffusivity}}{\text{thermal diffusivity}}$.

Question 34 [CRE — Reactor Design]

Bubble cap tray is:

Answer: (A)

Bubble cap trays have risers with caps for vapor distribution.

Question 35 [Fluid Mechanics — Pumps]

For CSTR, conversion X vs space time τ

Answer: (A)

For first order in CSTR: $X = \frac{k\tau}{1 + k\tau}$, not linear. For zero order: linear.

Question 36 [Process Calculations — Recycle]

Which are non-Newtonian fluids?

Answer:

(B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 37 [Mass Transfer — Absorption]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 38 [Heat Transfer — Heat Exchangers]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 39 [CRE — Reactor Design]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 40 [Fluid Mechanics — Non-Newtonian]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 41 [Process Calculations — Bypass]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 42 [Mass Transfer — Distillation]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 43 [Heat Transfer — Heat Exchangers]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 44 [CRE — Non-isothermal]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 45 [Fluid Mechanics — Laminar/Turbulent]

Which are non-Newtonian fluids?

Answer:

(B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 46 [Process Calculations — Material Balance]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 47 [Mass Transfer — Humidification]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 48 [Heat Transfer — Convection]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 49 [CRE — Homogeneous/Heterogeneous]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 50 [Fluid Mechanics — Reynolds Number]

Which are non-Newtonian fluids?

Answer: (B, C, D, E)

Most real fluids show non-Newtonian behavior to some degree.

Question 51 [Process Calculations — Material Balance]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time $\tau = V/Q$.

Answer: (10)

$\tau = V/Q = 100/10 = 10\text{ minutes}$.

Question 52 [Mass Transfer — Extraction]

Distillation: relative volatility $\alpha = 2.5$; Minimum reflux ratio at $x_D = 0.95$, $x_B = 0.1$.

Answer: (2)

For $\alpha = 2.5$, $R_m = (x_D/y_D) - 1 = 0.88/0.56 - 1 = 0.57$ (using Fenske).

Question 53 [Heat Transfer — Heat Exchangers]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time $\tau = V/Q$.

Answer: (10)

$\tau = V/Q = 100/10 = 10\text{ minutes}$.

Question 54 [CRE — CSTR/PFR]

Distillation: relative volatility $\alpha = 2.5$; Minimum reflux ratio at $x_D = 0.95$, $x_B = 0.1$.

Answer:

For ; $R_m = (x_{D/y_D}) - 1 \cdot \frac{H}{0.88/0.56} - 1 \cdot \frac{H}{0.57}$ (using Fenske).

Question 55 [Fluid Mechanics — Non-Newtonian]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time $\tau = \frac{V}{Q}$

Answer: (10)

$\tau = \frac{V}{Q} = 100/10 = 10\text{ minutes}$.

Question 56 [Process Calculations — Stoichiometry]

Distillation: relative volatility ; R_m Minimum reflux ratio at ; R_m " & α , β

Answer: (2)

For ; $R_m = (x_{D/y_D}) - 1 \cdot \frac{H}{0.88/0.56} - 1 \cdot \frac{H}{0.57}$ (using Fenske).

Question 57 [Mass Transfer — Distillation]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time $\tau = \frac{V}{Q}$

Answer: (10)

$\tau = \frac{V}{Q} = 100/10 = 10\text{ minutes}$.

Question 58 [Heat Transfer — Heat Exchangers]

Distillation: relative volatility ; R_m Minimum reflux ratio at ; R_m " & α , β

Answer: (2)

For ; $R_m = (x_{D/y_D}) - 1 \cdot \frac{H}{0.88/0.56} - 1 \cdot \frac{H}{0.57}$ (using Fenske).

Question 59 [CRE — Homogeneous/Heterogeneous]

CSTR: $V=100\text{L}$, $Q=10\text{ L/min}$. Space time $\tau = \frac{V}{Q}$

Answer: (10)

$\tau = \frac{V}{Q} = 100/10 = 10\text{ minutes}$.

Question 60 [Fluid Mechanics — Non-Newtonian]

Distillation: relative volatility ; R_m Minimum reflux ratio at ; R_m " & α , β

Answer: (2)

For ; $R_m = (x_{D/y_D}) - 1 \cdot \frac{H}{0.88/0.56} - 1 \cdot \frac{H}{0.57}$ (using Fenske).

